

ARIA HALAVATI

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EMPLOYMENT

Università Bocconi

Postdoc, funded by ERC Project ANGEVA

- Mentor: Antonio De Rosa

Milan, Italy

Sep 2025–Present

EDUCATION

Courant Institute of Mathematical Sciences, New York University

Ph.D. in Pure Mathematics

- Advisors: Guido De Philippis and Fang-Hua Lin
- Research area: Geometric measure theory and partial differential equations

New York, NY

Sep 2020–May 2025

Department of Mathematics, Sharif University of Technology

B.S. in Mathematics

- Advisor: Morteza Fotouhi

Tehran, Iran

2015–2019

PUBLICATIONS AND PREPRINTS

1. **Integral curvature estimates and geodesic limits for stable abelian Higgs fields.**
Marco Badran, Marco A.M. Guaraco, Aria Halavati. Preprint (2026), arXiv:2609.21893.
2. **Min-Max construction of anisotropic minimal surfaces with genus bound.**
Antonio De Rosa, Aria Halavati, Ling Wang. Preprint (2026), arXiv:2606.30098.
3. **Carleman estimates for stationary Q-valued maps: a variational approach.**
Aria Halavati, Luca Spolaor. Preprint (2025), arXiv:2508.14388.
4. **Decay of excess for the abelian Higgs model.**
Guido De Philippis, Aria Halavati, Alessandro Pigati. *J. Eur. Math. Soc.*, Online First (2026). doi:10.4171/JEMS/1819.
5. **Quantitative stability of Yang–Mills–Higgs instantons in two dimensions.**
Aria Halavati. *Arch. Ration. Mech. Anal.* **248**, Article 88 (2024). doi:10.1007/s00205-024-02035-2.
6. **New weighted inequalities on two-manifolds.**
Aria Halavati. *Rend. Sem. Mat. Univ. Padova*, Online First (2026). doi:10.4171/RSMUP/206.

INVITED TALKS

Westlake Winter School in Geometric Measure Theory Westlake University, Hangzhou, China; invited individual lecture (upcoming)	18–22 Jan 2027
Oberwolfach Workshop on Calculus of Variations Mathematisches Forschungsinstitut Oberwolfach, Germany; speaker	30 Aug–4 Sep 2026
Scuola Normale Superiore di Pisa Pisa, Italy	5–7 Nov 2025
UNAM Juriquilla, Querétaro, Mexico	6–8 Aug 2025
cSplash, Courant Institute New York, NY	5 Apr 2025
UT Austin Analysis Seminar Austin, TX	12 Mar 2025
Princeton DGGA Seminar Princeton, NJ	26 Feb 2025
MIT Analysis Seminar Boston, MA	12 Nov 2024
UC San Diego Geometry Seminar San Diego, CA	31 Oct 2024
Stanford Geometry Seminar Stanford, CA	16 Oct 2024
Virtual Analysis and PDE Seminar, hosted by UC Irvine Online	3 Oct 2024
Cornell Analysis Seminar Ithaca, NY	9 Sep 2024
Courant Geometric Analysis Seminar New York, NY	6 Sep 2024
University of Toronto Student PDE Seminar Online	14 Feb 2024

WORKSHOPS AND CONFERENCES

International geometric analysis conference in Milan II Bocconi University, Milan, Italy	8–12 Jun 2026
SLMath – Geometry and Analysis of Special Structures on Manifolds Berkeley, CA	18–22 Nov 2024
SLMath – Recent Progress on Geometric Analysis and Riemannian Geometry Berkeley, CA	21–25 Oct 2024
Joint Mathematics Meetings 2024 San Francisco, CA	3–6 Jan 2024
Oberwolfach PDE Workshop 2330 (Invited) Germany	23–29 Jul 2023

AWARDS AND HONORS

Peter Lax Fellowship , awarded to an outstanding Ph.D. student, Courant Institute	2022
First Prize , International Mathematical Competition, UCL	2017
Gold Medal , International Mathematical Olympiad	2015
Gold Medal , Iran's National Mathematical Olympiad	2014

ACADEMIC SERVICE

Reviewer for: *Canadian Journal of Mathematics*; *Journal of Functional Analysis*; *Annales de l'Institut Henri Poincaré*, *Analyse Non Linéaire*; *Adv. Math.*

Co-organized: Courant Graduate Student/Postdoc Seminar; social chapter of the Courant student organization; social climate survey administered for Courant student feedback; geometric measure theory reading group at Courant.

TEACHING

SLMath, Berkeley	17–28 Jul 2024
Teaching Assistant for the summer school <i>Special Geometric Structures and Analysis</i> , St. Mary's College	
• An introduction to geometric measure theory.	
Courant Institute of Mathematical Sciences, NYU. Teaching Assistant.	
Calculus I	Fall 2019; Fall 2021
Math for Econ	Spring 2020
Great Ideas in Math for non-math majors	Spring 2020
Linear Algebra	Fall 2022; Spring 2025
Probability and Statistics	Spring 2022
Analysis	Fall 2023; Spring 2024; Fall 2024
Young Scholars Club, Tehran, Iran. Tutoring Iran's IMO team	2016
Kerman, Iran. Math classes for students in my hometown; 7 national and 1 IMO medal from my class	2014–2015

SKILLS

Languages: Persian, English.

Programming: Python, TensorFlow, PyTorch, C++, MATLAB.

Music: Piano, Setar, Guitar, Oud.